

Highlights

DiracBench: A Reproducible Four-Method Framework for Relativistic Bound States with Scalar, Vector, and Tensor Interactions

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- Provides a reproducible four-method radial Dirac benchmark.
- Tests shooting, finite differences, Chebyshev, and DKB B-splines.
- Reports cross-method, residual, wavefunction, domain, and spurious-state diagnostics.
- Includes portable scripts, tests, machine-readable data, and publication figures.

DiracBench: A Reproducible Four-Method Framework for Relativistic Bound States with Scalar, Vector, and Tensor Interactions

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ARTICLE INFO

Keywords:

Dirac equation
radial eigenvalue problem
kinetic balance
B splines
Chebyshev collocation
finite differences
reproducibility

ABSTRACT

The release adds an explicit error-analysis table, automated regression tests, and a portable campaign driver so that the reported numerical agreement can be audited and regenerated outside the original development environment. We present DiracBench, a reproducible four-method framework for bound states of the spherical radial Dirac equation with scalar, vector, and radial tensor interactions. The implementation combines adaptive two-sided Runge–Kutta shooting, centered finite-difference matrix diagonalization, Chebyshev spectral collocation, and a dual-kinetic-balance (DKB) B-spline Galerkin discretization. Motivated by recent scalar–vector–tensor shooting calculations in coordinate space, but without reusing their fitted parameters or numerical results, we construct three independent dimensionless Woods–Saxon benchmark families and a controlled tensor-strength sweep. For the smooth benchmark P1, the four methods agree for the lowest states in the $\kappa = -1, 1, -2$ channels with maximum cross-method spreads of 5.3×10^{-6} , 1.22×10^{-5} , and 1.12×10^{-5} , respectively. Across $U_0 \in [-0.12, 0.12]$, the maximum four-method spread remains below 8.4×10^{-6} . A domain-lock study shows the shooting energy stable to approximately 10^{-10} once $r_{\max} \geq 20$ for P1. Phase-aligned common-grid comparisons give overlaps of 0.999990 and 0.999514 for finite differences and Chebyshev, respectively, relative to shooting. The centered finite-difference spectrum also exposes a grid-scale spurious state whose large component alternates sign at essentially every grid point, whereas the physical state has zero significant nodes. The DKB branch reaches few- 10^{-6} accuracy on the smooth benchmarks but exhibits non-monotonic basis refinement that remains a development target. The package contains the full numerical campaign, machine-readable CSV files, a dedicated error-analysis product, publication-quality figures, automated regression tests, pseudocode-linked LaTeX manuscript, and a Colab-ready notebook. These results establish an independent scalar–vector–tensor benchmark suite while retaining explicit diagnostics for unresolved discretization artifacts.

Program summary

Program title: DiracBench: Scalar–Vector–Tensor
Radial Dirac Benchmark Suite

CPC library link to program files: To be assigned
after acceptance.

Developer’s repository link: Not supplied in this
archive.

Licensing provisions: MIT.

Programming language: Python 3.

External routines/libraries: NumPy, SciPy, pandas,
Matplotlib, and pytest.

Nature of problem: Reproducible computation of
bound states of the spherical radial Dirac equation with
scalar, vector, and radial tensor interactions, including
the detection of discretization artifacts.

Solution method: Four independent numerical routes
are implemented: adaptive two-sided shooting, centered
finite-difference matrix diagonalization, Chebyshev col-
location, and a dual-kinetic-balance B-spline Galerkin
prototype.

Restrictions: The present campaign treats smooth
dimensionless Woods–Saxon potentials and selected κ
channels; the centered finite-difference branch retains
a documented grid-scale spurious mode, and the DKB
refinement study is not uniformly monotone.

Unusual features: The release couples cross-method
falsification, wavefunction overlap, residual and node di-
agnostics, domain locking, error-analysis tables, portable
campaign execution, automated tests, and publication-
quality figures.

1. Introduction

Reliable numerical solution of the Dirac equation re-
quires more than obtaining apparently stable eigenval-
ues. First-order relativistic operators admit discretiza-
tion artifacts that can masquerade as physical states,
and finite basis representations require a controlled
relationship between the large and small spinor com-
ponents. Dual kinetic balance (DKB) was introduced
to remove spurious states and improve convergence
in finite-basis Dirac calculations [1, 8]. Kinematically
balanced B-spline approaches have subsequently been
used successfully in relativistic eigenvalue problems
[3, 4, 6, 7]. More recently, Li *et al.* showed explicitly

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that centered finite differences on a uniform radial grid can generate spurious states and proposed a staggered-grid scheme that suppresses the artifact [2, 9, 25].

A recent coordinate-space solver by Kiessling *et al.* applies two-sided Runge–Kutta shooting to spherical scalar, vector, and tensor Woods–Saxon potentials and discusses practical sensitivity to state identification and numerical implementation [5, 9, 11, 16]. We use only these methodological ideas: no fitted parameters, energy tables, or wavefunctions from that work are reproduced here. Instead, all scalar–vector–tensor data below are generated from independent dimensionless benchmark sets defined specifically for DiracBench.

DiracBench is designed around cross-method falsification: a state is accepted only when independent discretizations converge to the same physical solution and pass residual, normalization, localization, node, and domain-size tests. CPC software studies have established useful precedents for radial Dirac solvers, B-spline programs, finite-element atomic calculations, partial-wave codes, and reusable relativistic software [10, 12, 13, 14, 17, 20, 18, 15]. This manuscript documents the first reproducible prototype and, equally importantly, identifies the numerical branch that has not yet reached production quality.

2. Radial Dirac formulation and common conventions

We begin from the stationary Dirac equation

$$H\Psi(\mathbf{r}) = E\Psi(\mathbf{r}),$$

with

$$H = \boldsymbol{\alpha} \cdot \mathbf{p} + \beta[m + S(r)] + V(r) + H_T.$$

In the standard representation,

$$\boldsymbol{\alpha} = \begin{pmatrix} 0 & \boldsymbol{\sigma} \\ \boldsymbol{\sigma} & 0 \end{pmatrix}, \quad \beta = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}.$$

For spherical symmetry we use

$$\Psi_{\kappa m}(\mathbf{r}) = \frac{1}{r} \begin{pmatrix} F(r)\Omega_{\kappa m} \\ iG(r)\Omega_{-\kappa m} \end{pmatrix},$$

together with

$$(\boldsymbol{\sigma} \cdot \hat{\mathbf{r}})\Omega_{\kappa m} = -\Omega_{-\kappa m}, \quad (\boldsymbol{\sigma} \cdot \mathbf{L} + 1)\Omega_{\kappa m} = -\kappa\Omega_{\kappa m}. \quad (5)$$

Using

$$\boldsymbol{\sigma} \cdot \mathbf{p} = -i(\boldsymbol{\sigma} \cdot \hat{\mathbf{r}}) \left(\frac{d}{dr} + \frac{1}{r} - \frac{\boldsymbol{\sigma} \cdot \mathbf{L}}{r} \right), \quad (6)$$

one obtains

$$\boldsymbol{\sigma} \cdot \mathbf{p} \left[\frac{F}{r} \Omega_{\kappa m} \right] = \frac{i}{r} \left(\frac{dF}{dr} + \frac{\kappa}{r} F \right) \Omega_{-\kappa m}, \quad (7)$$

$$\boldsymbol{\sigma} \cdot \mathbf{p} \left[\frac{G}{r} \Omega_{-\kappa m} \right] = \frac{i}{r} \left(\frac{dG}{dr} - \frac{\kappa}{r} G \right) \Omega_{\kappa m}. \quad (8)$$

For the radial tensor convention implemented in DiracBench,

$$K_{\kappa}(r) = \frac{\kappa}{r} - U(r). \quad (9)$$

Introducing

$$\Sigma(r) = V(r) + S(r), \quad \Delta(r) = V(r) - S(r), \quad (10)$$

the radial equations are

$$\left(\frac{d}{dr} + K_{\kappa} \right) F = [E + m - \Delta(r)]G, \quad (11)$$

$$\left(\frac{d}{dr} - K_{\kappa} \right) G = -[E - m - \Sigma(r)]F. \quad (12)$$

Equivalently,

$$\frac{dF}{dr} = -K_{\kappa}F + [E + m - \Delta(r)]G, \quad (13)$$

$$\frac{dG}{dr} = K_{\kappa}G - [E - m - \Sigma(r)]F. \quad (14)$$

The common matrix form is

$$\begin{pmatrix} m + \Sigma & -d/dr + K_{\kappa} \\ d/dr + K_{\kappa} & -m + \Delta \end{pmatrix} \begin{pmatrix} F \\ G \end{pmatrix} = E \begin{pmatrix} F \\ G \end{pmatrix}. \quad (15)$$

2.1. Boundary behaviour

For a point Coulomb potential $V = -Z\alpha/r$, the regular solution behaves as

$$F(r) \propto r^{\gamma}, \quad \gamma = \sqrt{\kappa^2 - (Z\alpha)^2}, \quad (16)$$

and the implemented convention gives

$$\frac{G}{F} = \frac{\gamma + \kappa}{Z\alpha}. \quad (17)$$

For a bound state at large radius,

$$\lambda = \sqrt{m^2 - E^2}, \quad F, G \propto e^{-\lambda r}, \quad \frac{G}{F} \sim -\frac{\lambda}{E + m}. \quad (18)$$

3. Numerical methods: integrated derivation, discretization, and pseudocode

Each solver below is derived directly from the same radial operator. The derivation and algorithm are intentionally kept together so that every computational step can be traced to the underlying equations, following standard treatments of relativistic atomic structure and numerical differential equations [27, 26, 28].

Algorithm 1 Adaptive two-sided shooting for one radial Dirac state

Require: $S, V, U, \kappa, [r_{\min}, r_{\max}]$, energy bracket $[E_a, E_b]$

Ensure: Eigenvalue E and normalized radial spinor (F, G)

- 1: Choose matching radius r_m
 - 2: **for** each trial energy E requested by the root finder **do**
 - 3: Construct regular inner boundary data at $r_{\min}$
 - 4: Integrate Eqs. (13)–(14) outward to r_m
 - 5: Construct exponentially decaying outer data at $r_{\max}$
 - 6: Integrate inward from $r_{\max}$ to r_m
 - 7: Evaluate $D(E) = F_{\text{out}}G_{\text{in}} - G_{\text{out}}F_{\text{in}}$
 - 8: **end for**
 - 9: Root-find $D(E) = 0$ inside the bracket
 - 10: Reintegrate, rescale the inward branch, join, and normalize
 - 11: **return** E, F, G
-

Algorithm 2 Finite-difference eigensolver with spurious-state diagnostics

Require: $S, V, U, \kappa, r_{\max}, N$, target E_t

Ensure: Ranked physical-state candidates

- 1: Set $h = r_{\max}/(N + 1)$; construct r_i and centered derivative matrix D
 - 2: Build diagonal matrices A, B , and K
 - 3: Assemble the $2N \times 2N$ block Hamiltonian from A, B, D , and K
 - 4: Compute several shift-invert eigenpairs near E_t
 - 5: **for** each candidate (E_j, F_j, G_j) **do**
 - 6: Normalize; compute residual, nodes, oscillation score, and tail weight
 - 7: Check stability under grid and domain refinement
 - 8: **end for**
 - 9: Flag grid-scale oscillatory or nonconvergent candidates
 - 10: **return** ranked candidates
-

3.1. Adaptive two-sided shooting and Runge–Kutta integration

With

$$\mathbf{y} = (F, G)^T, \quad (19)$$

Eqs. (13)–(14) become

$$\frac{d\mathbf{y}}{dr} = A(r; E)\mathbf{y}, \quad (20)$$

where

$$A = \begin{pmatrix} -K_\kappa & E + m - \Delta \\ -[E - m - \Sigma] & K_\kappa \end{pmatrix}. \quad (21)$$

An outward solution and an inward solution represent the same eigenstate at $r = r_m$ when they are linearly dependent. Hence

$$D(E) = \det \begin{pmatrix} F_{\text{out}} & F_{\text{in}} \\ G_{\text{out}} & G_{\text{in}} \end{pmatrix}_{r_m} = F_{\text{out}}G_{\text{in}} - G_{\text{out}}F_{\text{in}} = 0, \quad (22)$$

DiracBench finds the root of this secular function and then rescales and joins the two integrations. The use of explicit bracketing, adaptive integration, and domain refinement follows the convergence-oriented practice of open radial Dirac solvers [10, 28].

3.2. Finite-difference matrix diagonalization

For

$$r_i = ih, \quad h = \frac{r_{\max}}{N + 1}, \quad (23)$$

the centered derivative is

$$f'(r_i) \approx \frac{f_{i+1} - f_{i-1}}{2h}. \quad (24)$$

Thus

$$D_{i,i+1} = \frac{1}{2h}, \quad D_{i,i-1} = -\frac{1}{2h}. \quad (25)$$

Defining

$$A = \text{diag}[m + \Sigma(r_i)], \quad (26)$$

$$B = \text{diag}[-m + \Delta(r_i)], \quad (27)$$

$$K = \text{diag}[K_\kappa(r_i)], \quad (28)$$

gives

$$H_{\text{FD}} = \begin{pmatrix} A & -D + K \\ D + K & B \end{pmatrix}. \quad (29)$$

The centered derivative has lattice symbol

$$\tilde{D}(k) = \frac{i \sin(kh)}{h}, \quad (30)$$

which vanishes at both $k = 0$ and $k = \pi/h$. This extra zero explains the fermion-doubling/spurious-mode risk and motivates a staggered-grid production branch, consistent with the broader coordinate-space literature on suppressing fermion doubling [9, 25].

3.3. Chebyshev spectral collocation

The Chebyshev–Lobatto nodes are

$$x_j = \cos \frac{j\pi}{N}, \quad j = 0, \dots, N, \quad (31)$$

mapped by

$$r = \frac{r_{\max}}{2}(1 - x). \quad (32)$$

Therefore

$$\frac{d}{dr} = -\frac{2}{r_{\max}} \frac{d}{dx}, \quad D_r = -\frac{2}{r_{\max}} D_x. \quad (33)$$

Algorithm 3 Chebyshev collocation solver**Require:** $S, V, U, \kappa, r_{\max}, N$, target E_t **Ensure:** Collocation eigenpair and diagnostics

- 1: Form Lobatto nodes $x_j = \cos(j\pi/N)$ and map to r_j
- 2: Build D_x and set $D_r = -2D_x/r_{\max}$
- 3: Remove endpoint degrees of freedom
- 4: Evaluate potentials and assemble H_{Ch}
- 5: Diagonalize the dense collocation matrix
- 6: Reject eigenvalues with significant imaginary parts
- 7: Rank remaining states by target proximity, residual, and oscillation
- 8: **return** best physical candidate

Algorithm 4 Dual-kinetic-balance B-spline Galerkin solver**Require:** S, V, U, κ , knots, spline degree p , basis size N_B **Ensure:** Generalized eigenpairs and diagnostics

- 1: Construct endpoint-compatible B-splines B_i and their derivatives
- 2: **for** each B_i **do**
- 3: Form the balanced spinors u_i and v_i
- 4: **end for**
- 5: Select radial quadrature and evaluate H_{ij} and S_{ij}
- 6: Symmetrize roundoff in H and S
- 7: Solve $HC = ESC$
- 8: Reconstruct and normalize candidate spinors
- 9: Evaluate residual, nodes, and basis/domain refinement
- 10: **return** physical candidates

128 After eliminating endpoint degrees of freedom,

$$H_{\text{Ch}} = \begin{pmatrix} A & -D_r + K \\ D_r + K & B \end{pmatrix}. \quad (34)$$

129 This global differentiation operator is independent of
 130 the local finite-difference stencil and therefore provides
 131 a valuable cross-check. The Chebyshev construction
 132 follows the standard spectral-collocation framework
 133 [29].

3.4. Dual-kinetic-balance B-spline Galerkin discretization

134 From Eq. (11),

$$G = \frac{1}{E + m - \Delta} \left(\frac{d}{dr} + K_\kappa \right) F. \quad (35)$$

137 In the positive-energy nonrelativistic limit $E = m + \epsilon$
 138 with $|\epsilon| \ll m$ and weak potentials,

$$G \approx \frac{1}{2m} \left(\frac{d}{dr} + K_\kappa \right) F. \quad (36)$$

139 A kinetically balanced basis therefore couples the small-
 140 component basis to the differentiated large-component
 141 basis. This relation is the central variational safeguard
 142 behind kinetic-balance and DKB constructions [21, 23,
 143 24, 8]. The DKB prototype uses

$$\begin{aligned} u_i &= \begin{pmatrix} B_i \\ \frac{1}{2m} \left(\frac{d}{dr} + K_\kappa \right) B_i \end{pmatrix}, \\ v_i &= \begin{pmatrix} \frac{1}{2m} \left(-\frac{d}{dr} + K_\kappa \right) B_i \\ B_i \end{pmatrix}. \end{aligned} \quad (37)$$

144 With

$$\Phi = \sum_i c_i u_i + \sum_i d_i v_i, \quad (38)$$

145 Galerkin projection yields

$$HC = ESC, \quad (39)$$

146 where

$$H_{ij} = \int \Phi_i^\dagger H_\kappa \Phi_j dr, \quad S_{ij} = \int \Phi_i^\dagger \Phi_j dr. \quad (40)$$

4. Validation protocol and mathematical diagnostics

4.1. Residuals, normalization, and cross-method wavefunction agreement

For a numerical spinor,

$$\mathbf{R} = (H_\kappa - E) \begin{pmatrix} F \\ G \end{pmatrix}, \quad (41)$$

with

$$R_F = (m + \Sigma - E)F - \frac{dG}{dr} + K_\kappa G, \quad (42)$$

$$R_G = \frac{dF}{dr} + K_\kappa F + (-m + \Delta - E)G. \quad (43)$$

The reported normalized residual is

$$\mathcal{R} = \frac{[\int (R_F^2 + R_G^2) dr]^{1/2}}{[\int (F^2 + G^2) dr]^{1/2}}. \quad (44)$$

After normalization,

$$\int (F^2 + G^2) dr = 1. \quad (45)$$

Residual, overlap, and refinement checks are used together because variational or Galerkin approximations of operators with a spectral gap can exhibit spectral pollution even when isolated eigenvalues appear numerically stable [22, 25]. For two independently computed real spinors, define

$$\mathcal{O}_{ij} = \int (F_i F_j + G_i G_j) dr, \quad \sigma_{ij} = \text{sgn}(\mathcal{O}_{ij}), \quad (46)$$

and

$$\Delta\psi_{ij} = \left[\int (F_i - \sigma_{ij} F_j)^2 + (G_i - \sigma_{ij} G_j)^2 dr \right]^{1/2}. \quad (47)$$

Algorithm 5 Cross-method validation of an identified physical state

Require: Solutions from two or more independent solvers

Ensure: Energy, spinor, residual, and domain-agreement diagnostics

```

1: Interpolate all spinors to a common radial mesh
   and normalize
2: for each method pair  $(i, j)$  do
3:   Compute  $\Delta E_{ij} = |E_i - E_j|$ 
4:   Compute signed overlap and choose the optimal
   global sign
5:   Compute phase-aligned  $\Delta\psi_{ij}$ 
6: end for
7: for each method  $i$  do
8:   Evaluate residual  $\mathcal{R}_i$ , nodes, oscillation, and tail
   localization
9:   Repeat after increasing resolution and  $r_{\max}$ 
10: end for
11: Accept only states whose physical diagnostics con-
   verge consistently

```

4.2. Exact Dirac–Coulomb oracle and its derivation

For

$$V = -\frac{Z\alpha}{r}, \quad S = U = 0, \quad (48)$$

define

$$\gamma = \sqrt{\kappa^2 - (Z\alpha)^2}, \quad n_r = n - |\kappa|. \quad (49)$$

The exact quantization condition is

$$\frac{Z\alpha E}{\sqrt{m^2 - E^2}} = n_r + \gamma. \quad (50)$$

Squaring,

$$(Z\alpha)^2 E^2 = (n_r + \gamma)^2 (m^2 - E^2), \quad (51)$$

so

$$E^2 [(Z\alpha)^2 + (n_r + \gamma)^2] = m^2 (n_r + \gamma)^2. \quad (52)$$

Therefore

$$E_{n\kappa} = m \left[1 + \frac{(Z\alpha)^2}{(n - |\kappa| + \sqrt{\kappa^2 - (Z\alpha)^2})^2} \right]^{-1/2}. \quad (53)$$

For $n = 1$, $\kappa = -1$, $Z\alpha = 0.2$, and $m = 1$,

$$E = \sqrt{1 - 0.2^2} = 0.9797958971132712. \quad (54)$$

This analytic oracle follows the standard Dirac–Coulomb spectrum used in relativistic atomic-structure calculations [26, 27].

Table 1

Original DiracBench smooth benchmark sets.

Set	S_0	V_0	U_0	R	a
P1	-0.60	0.20	0.00	5.0	0.60
P2	-0.60	0.20	0.08	5.0	0.60
P3	-0.70	0.25	-0.06	5.5	0.75

5. Original smooth scalar–vector–tensor benchmark design

We define the independent Woods–Saxon family

$$f(r; R, a) = \left[1 + \exp\left(\frac{r - R}{a}\right) \right]^{-1}, \quad (55)$$

$$S(r) = S_0 f, \quad V(r) = V_0 f, \quad U(r) = U_0 f. \quad (56)$$

All quantities in this campaign are dimensionless with $m = 1$. The three parameter sets in Table 1 span a baseline well, a positive tensor perturbation, and a deeper/more diffuse well. They are not fits to any nucleus and are not copied from published numerical tables. For each matrix method, the requested state is identified in the same κ channel and tracked under refinement. Shift-invert targeting locates the relevant part of the spectrum but does not modify the independently assembled Hamiltonian.

6. Four-method energy agreement

The largest spread is 1.22×10^{-5} . The three independent potential families give similarly small spreads: Figure 1 visualizes both the energies and the absolute deviations from the independent shooting reference. The paired view makes the agreement scale visible without relying on rounded table entries.

7. Error analysis and numerical reliability

The release separates energy error, method-to-method spread, wavefunction agreement, domain truncation, and oscillatory-state diagnostics. For a matrix-method energy E_i and the independently computed shooting energy E_s , we report $|E_i - E_s|$ and the relative error $|E_i - E_s|/|E_s|$. The cross-method spread is the maximum minus the minimum of the four energies for one identified state; it is a numerical-agreement measure rather than a statistical uncertainty.

For common-grid spinors, the global sign is chosen to make the overlap positive before computing the two-component L^2 difference. Domain error compares each shooting result with the largest tested domain, $r_{\max} = 35$. Finally, the centered-FD alternation fraction records the proportion of adjacent large-component samples with opposite signs and is used

Table 2

Four-method energies for P1 at $r_{\max} = 25$. FD uses $N = 900$, Chebyshev uses $N = 120$, and DKB uses 46 B-splines of degree 5.

κ	Shooting	FD	Chebyshev	DKB B-spline	spread
-1	0.7985749864	0.7985708513	0.7985749868	0.7985697115	5.28×10^{-6}
1	0.9725652688	0.9725550503	0.9725636160	0.9725672851	1.22×10^{-5}
-2	0.9285358974	0.9285331465	0.9285359236	0.9285247109	1.12×10^{-5}

Table 3

Lowest $\kappa = -1$ energies for the original parameter sets at $r_{\max} = 30$.

Set	Shooting	FD	Chebyshev	DKB B-spline	spread
P1	0.7985749864	0.7985701621	0.7985749865	0.7985709189	4.82×10^{-6}
P2	0.8258226101	0.8258186587	0.8258226101	0.8258190139	3.95×10^{-6}
P3	0.7327349195	0.7327303606	0.7327349195	0.7327273352	7.58×10^{-6}

together with node counts and cross-method disagree-
ment to identify the grid-scale branch. All rows are
written to `data/error_analysis.csv`, with aggregate
extrema in `data/error_analysis_summary.json`.

8. Tensor-strength sweep

Holding $S_0 = -0.60$, $V_0 = 0.20$, $R = 5$, and $a = 0.60$
fixed while varying U_0 from -0.12 to 0.12 produces the
monotonic shift in Fig. 4. The maximum four-method
spread over the complete sweep is 8.39×10^{-6} .

9. Domain-size and discretization refinement

Figure 5 varies $r_{\max}$ from 15 to 35 while main-
taining approximately constant FD grid density and

DKB basis density. The shooting energy changes from
 0.7985749313 at $r_{\max} = 15$ to 0.7985749863 at $r_{\max} =$
 20 , and is then stable at approximately the 10^{-10}
level. Chebyshev follows the same limit closely. At
fixed $r_{\max} = 25$, the FD energy error decreases from
 8.31×10^{-5} at $N = 200$ to 2.33×10^{-6} at $N = 1200$.
Chebyshev falls below 2×10^{-9} by $N = 60$. The DKB
calculation improves sharply between 28 and 34 basis
functions but then fluctuates at the few- 10^{-6} level
rather than converging monotonically; this plateau is
retained explicitly.

10. Wavefunction agreement

After common-grid renormalization and phase align-
ment, the FD-shooting overlap is 0.99998968 with

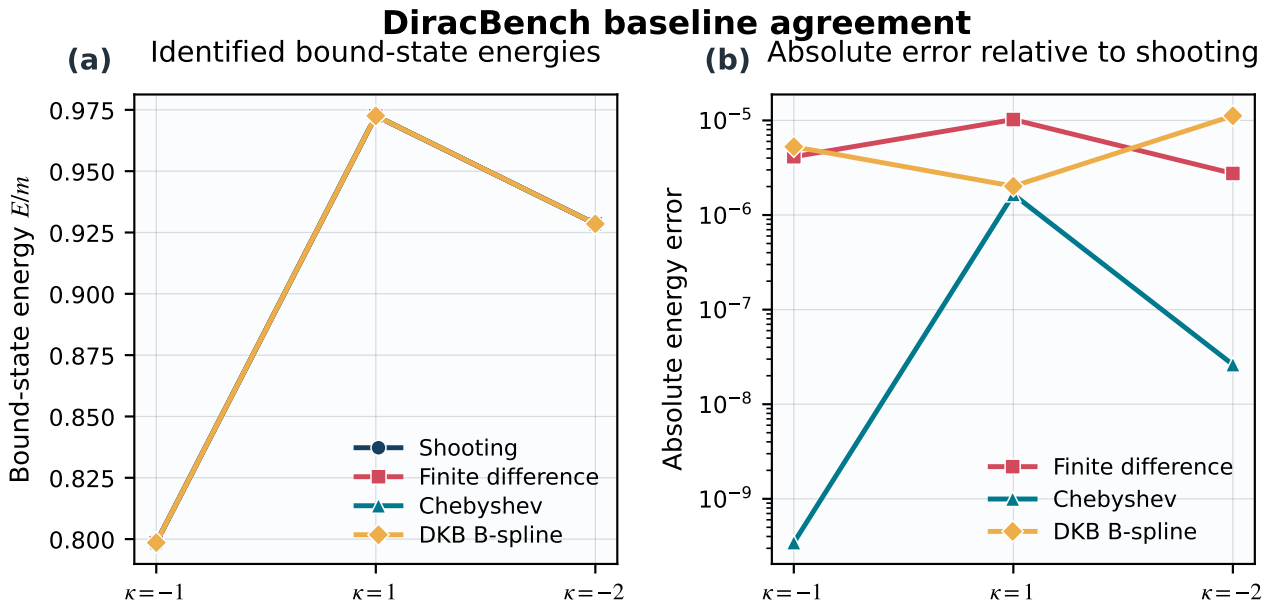


Figure 1: Baseline four-method agreement for the three angular channels. The left panel shows the identified energies; the right panel shows absolute deviations from two-sided shooting on a logarithmic scale.

DiracBench error and convergence dashboard

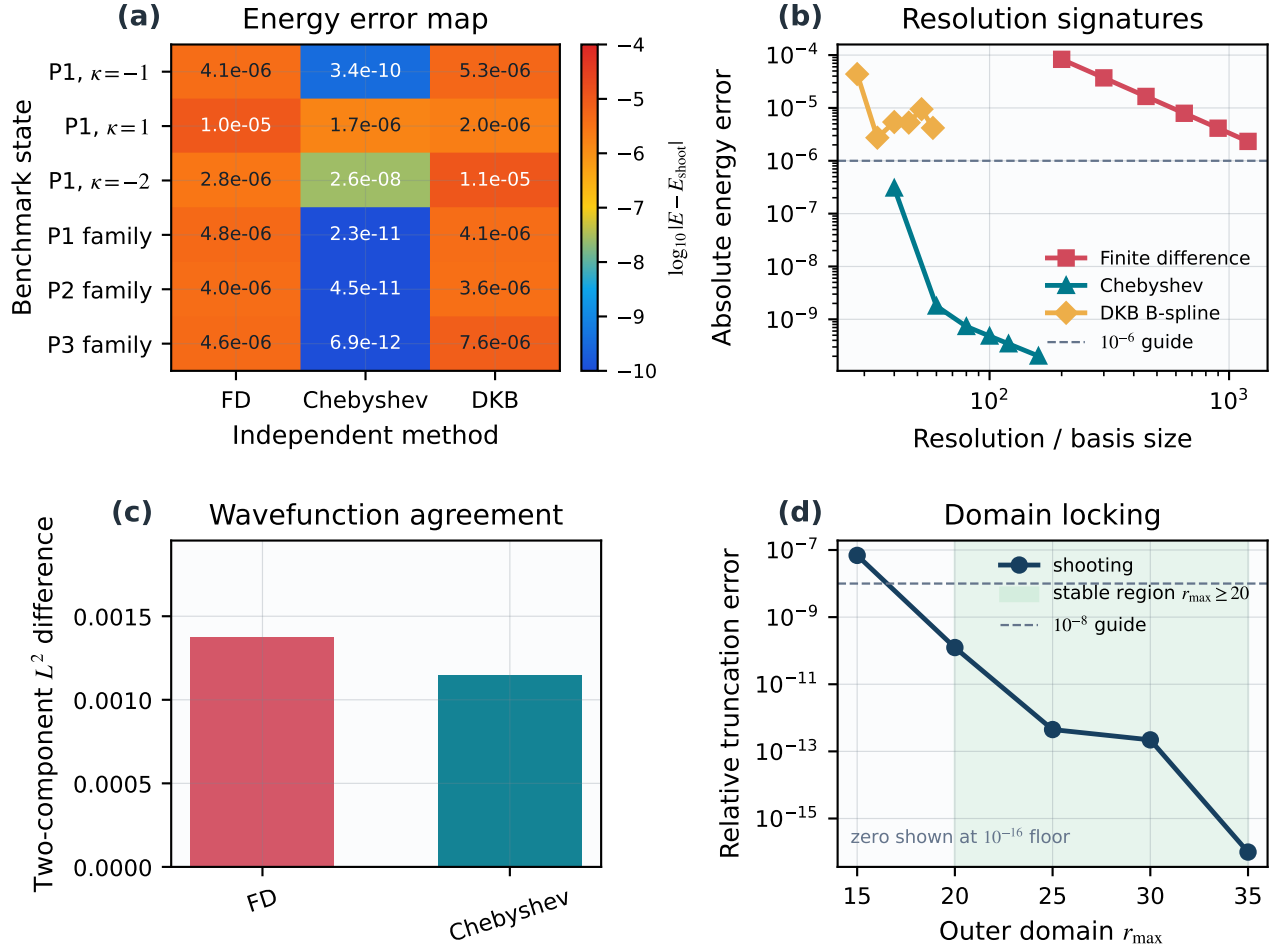
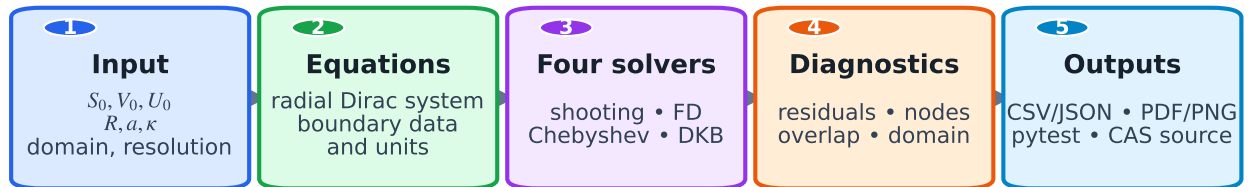


Figure 2: Error-analysis dashboard. (a) Annotated logarithmic energy-error map relative to shooting; (b) resolution signatures; (c) common-grid wavefunction differences, with FD and Chebyshev overlaps of 0.999990 and 0.999514, respectively; and (d) outer-domain locking. The diagnostics are complementary and are not interpreted as a single probabilistic error bar.

DiracBench reproducibility workflow



one benchmark specification → four independent numerical branches → auditable release products

Figure 3: Reproducible DiracBench workflow. A single benchmark specification is sent independently to the four numerical branches; the resulting energies and spinors are then checked by residual, normalization, node, overlap, and domain diagnostics before machine-readable data, vector figures, tests, and manuscript-ready outputs are written.

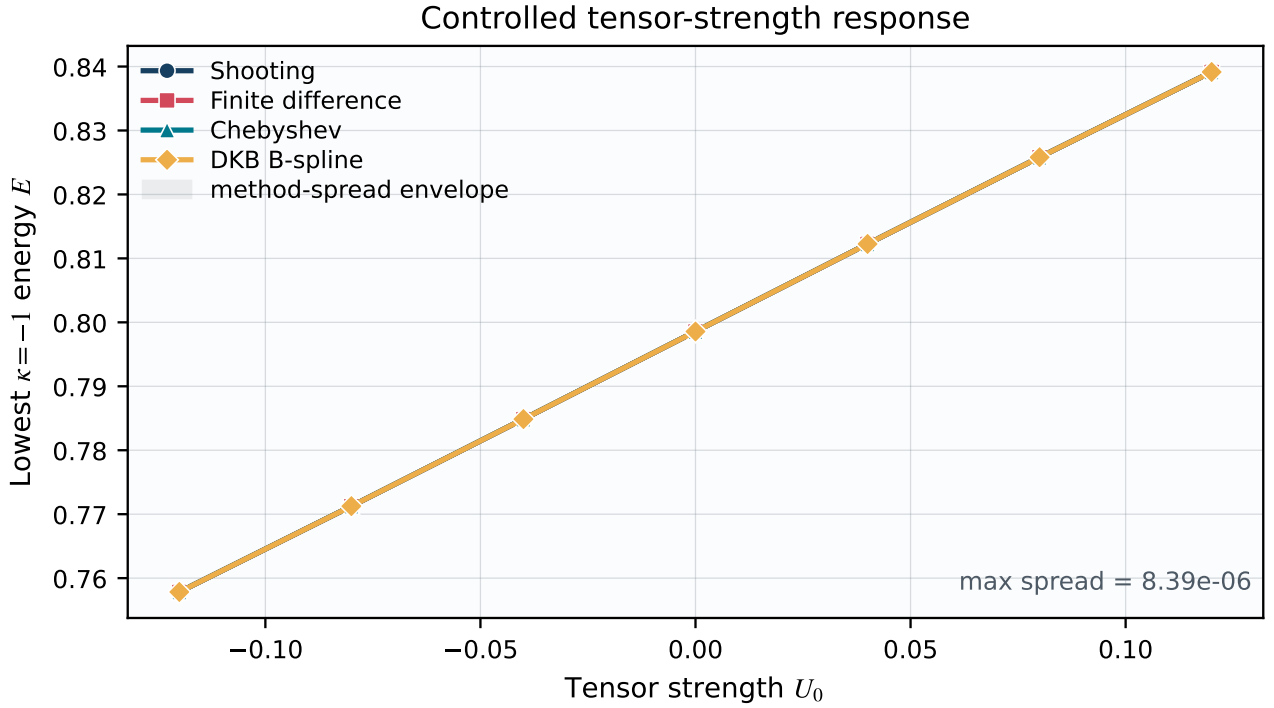


Figure 4: Tensor-strength response for P1. The shaded band is the four-method energy-spread envelope.

two-component L^2 difference 1.38×10^{-3} , while the Chebyshev–shooting overlap is 0.99951424 with L^2 difference 1.15×10^{-3} . These are global two-component diagnostics; the energy agreement is substantially tighter than the pointwise wavefunction difference.

shooting and Chebyshev spectra. These tests are especially important because centered differences can generate spurious Dirac states [2, 9, 25]. A future release will add a staggered-grid formulation while retaining the centered scheme as a reproducible counterexample.

11. Node and spurious-state diagnostics

At $N = 900$ in the P1 $\kappa = -1$ centered-FD calculation, the physical candidate at $E = 0.7985708513$ has zero significant nodes and zero alternation fraction. A second positive candidate at $E = 0.9725550503$ has 789 sign changes above a 1% amplitude threshold and an alternation fraction of 1.0. Shooting and Chebyshev do not identify a corresponding physical bound state in this channel.

12. Performance benchmark

Median wall-clock times over three repetitions in the present Python environment are 0.0865 s for shooting, 0.0083 s for FD, 0.0316 s for Chebyshev, and 0.1440 s for DKB. These values are hardware- and implementation-dependent and are reported only as a reproducible baseline.

13. Additional spurious-state safeguards

The explicit diagnostic above is supplemented by refinement checks and comparison with independent

14. Reproducibility and software

The archive contains the reusable Python solver module, a portable campaign driver, a Colab-ready notebook, CSV tables for every numerical sweep, a machine-readable error-analysis table, publication-quality PDF/PNG figures, automated regression tests, dependency metadata, an MIT license, and the present $\text{\LaTeX}$ source. The Python implementation uses the array, scientific-computing, plotting, and notebook ecosystems described by Harris *et al.*, Virtanen *et al.*, Hunter, and Kluyver *et al.* [30, 31, 32, 35]. The reproducibility record follows established recommendations for computational research and FAIR data stewardship [33, 34], while the package-oriented release is consistent with CPC’s open-source software tradition [19]. The command

```
python run_campaign.py --output generated_campaign
```

regenerates numerical outputs, error analysis, metadata, and figures without relying on a hard-coded path or deleting an existing output directory. All manuscript figures are regenerated from machine-readable outputs.

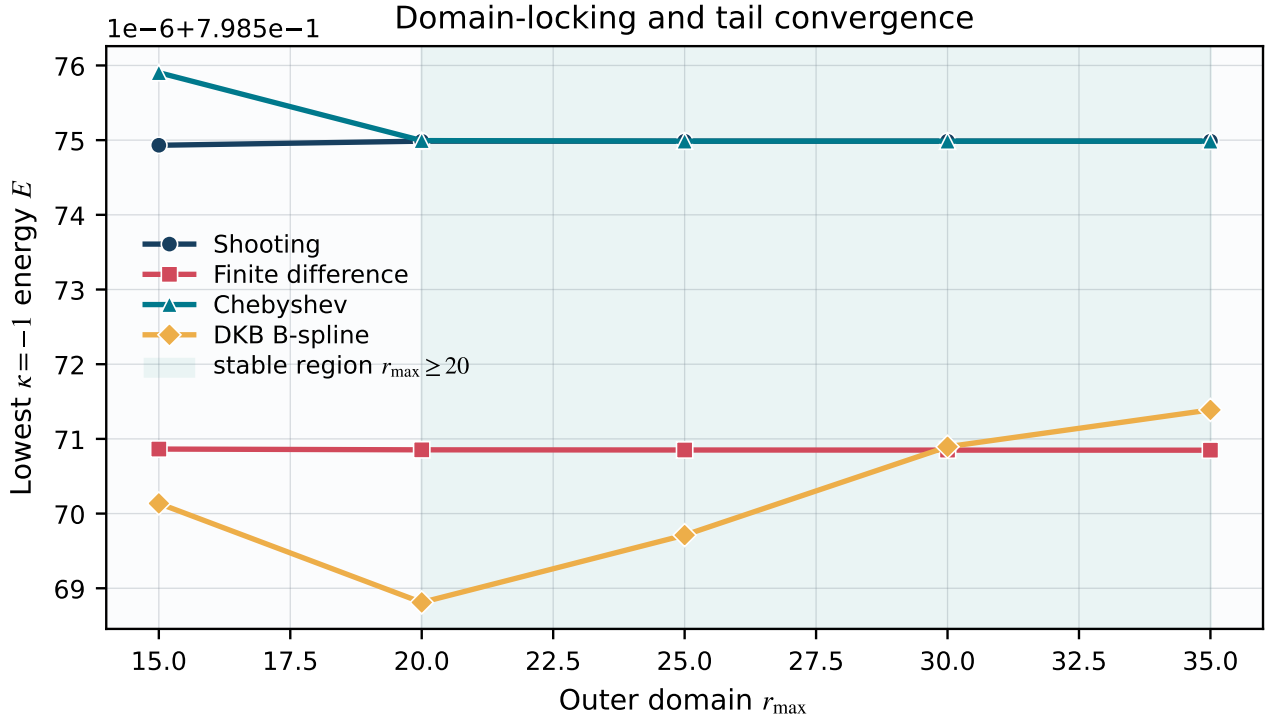


Figure 5: $r_{\max}$ -independence test for the lowest P1 $\kappa = -1$ state. The highlighted region marks the domain range used for the stability claim.

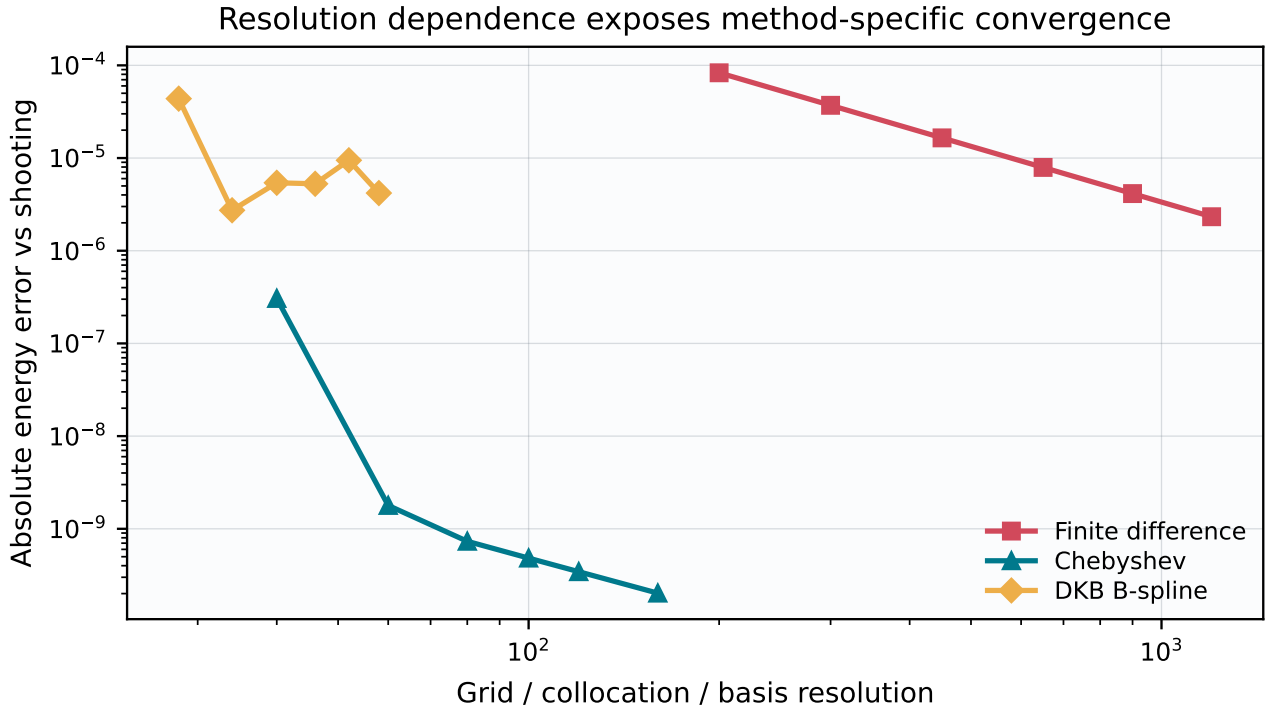


Figure 6: Resolution study for P1, $\kappa = -1$, relative to the converged shooting energy.

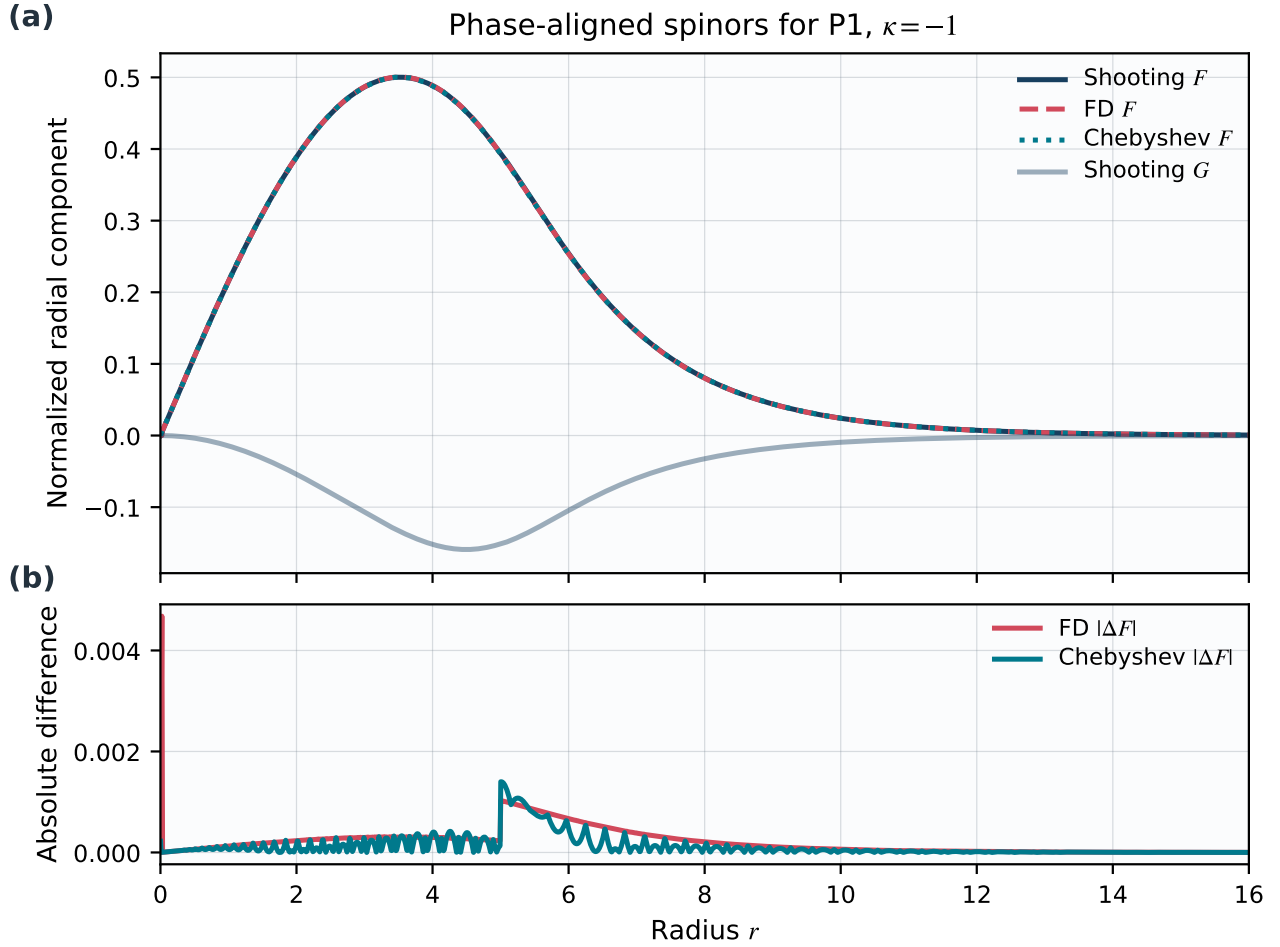


Figure 7: Phase-aligned two-component wavefunction comparison for P1, $\kappa = -1$. The lower panel shows absolute differences in the large component relative to shooting.

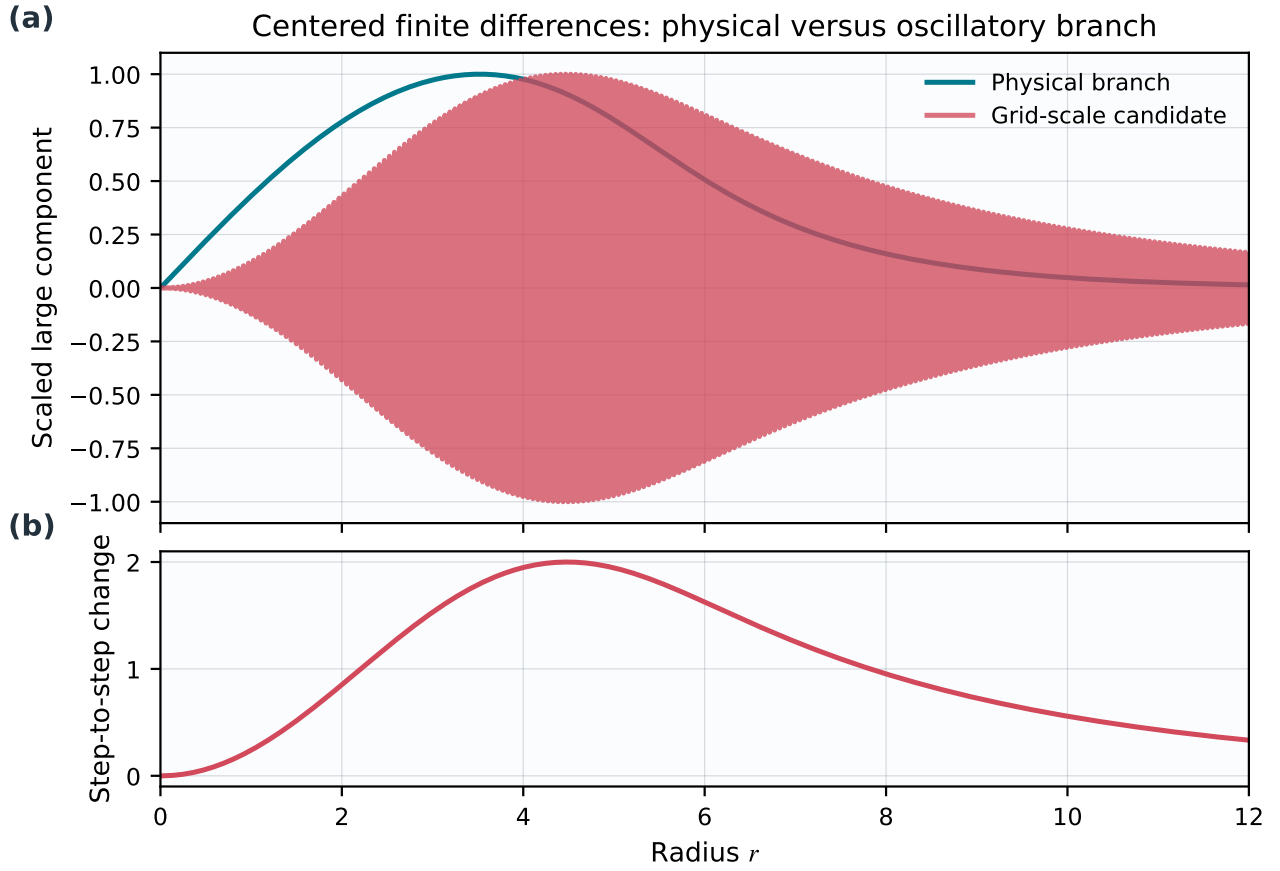


Figure 8: Centered-FD physical state and grid-scale oscillatory candidate for P1, $\kappa = -1$. The lower panel displays the step-to-step change that exposes the alternating branch.

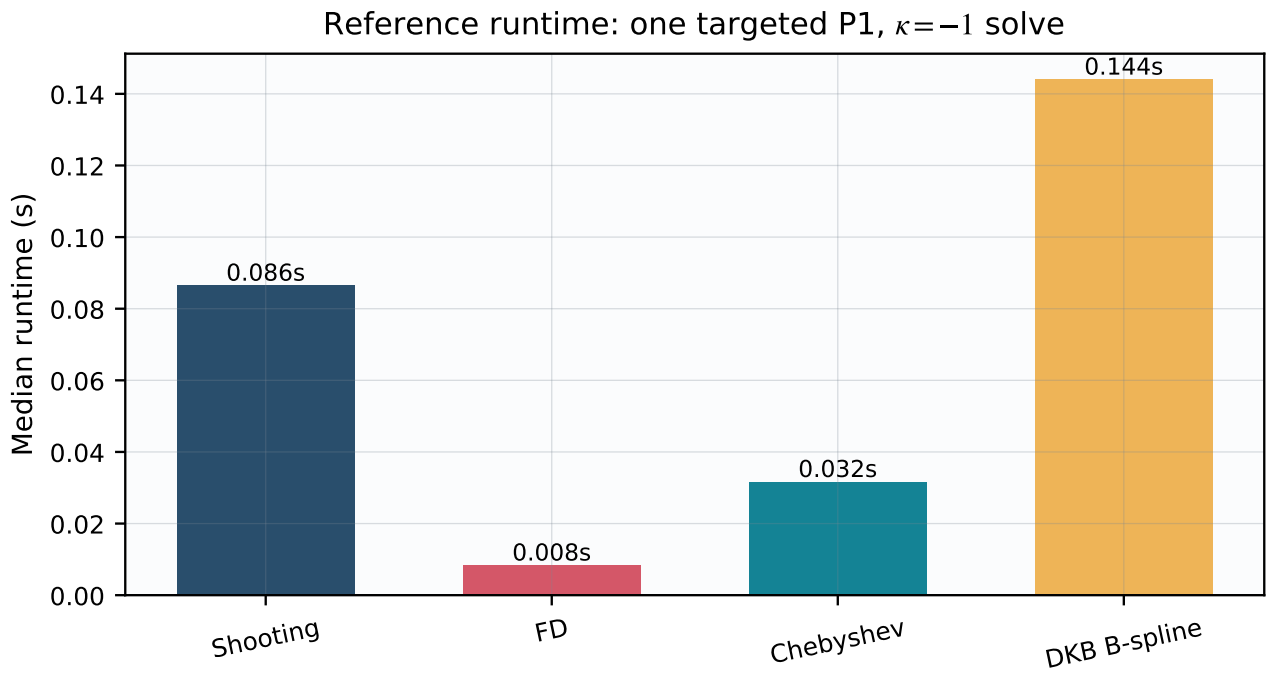


Figure 9: Median runtime baseline for one targeted P1 $\kappa = -1$ solve.

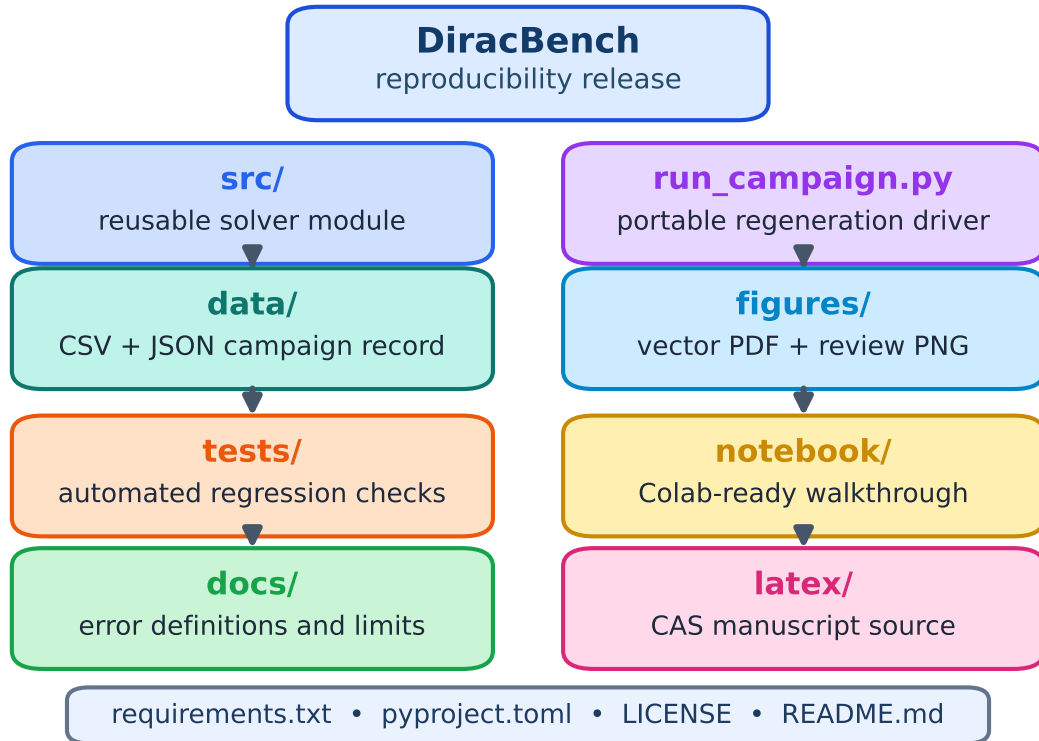


Figure 10: Archive structure of the reproducibility release. Source code and the portable campaign driver generate CSV/JSON data and publication figures; tests, dependency metadata, the notebook, and the CAS manuscript remain visible as separately inspectable release components.

15. Limitations and next production stage

The smooth-potential campaign establishes four-method agreement at the 10^{-5} scale or better for the tested states, but two limitations remain. First, the centered FD method still possesses the expected doubling artifact; its physical branch converges cleanly, while the oscillatory branch is retained as a reproducible diagnostic. Second, the DKB branch is accurate at the few- 10^{-6} level on the smooth benchmarks yet its basis-refinement curve is non-monotonic. The current release therefore reports conditioning and overlap diagnostics rather than presenting DKB as uniformly production-ready. A staggered-grid alternative, improved knot placement, and additional singular-potential validation are natural next steps. The singular point-Coulomb DKB benchmark remains separate and is not used to support the smooth-potential four-method claim.

16. Conclusions

DiracBench now moves beyond a prototype plan to an original numerical scalar-vector-tensor campaign. Three independently chosen smooth Woods-Saxon parameter sets, three angular channels, a tensor-strength sweep, domain locking, resolution refinement, wave-function comparison, node diagnostics, and runtime measurements have all been generated by the present code. For the principal smooth tests, the four methods

identify the same bound states with cross-method energy spreads below 1.3×10^{-5} , and the tensor sweep remains below 8.4×10^{-6} . Shooting and Chebyshev show near-identical energies, FD converges systematically, and DKB reaches few- 10^{-6} accuracy with a non-monotonic basis plateau that is reported rather than hidden. The centered FD calculation also produces a clear grid-scale oscillatory state, demonstrating the practical value of node and cross-method diagnostics.

Data and code availability

All source code, dependency metadata, MIT license, automated tests, the Colab notebook, CSV outputs, error-analysis products, publication figures, and figure-generation scripts used for the scalar-vector-tensor campaign are included in the accompanying reproducibility archive. A public repository and archival DOI should be added to the final submission record so that the exact release can be cited by CPC.

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